ORDER TRENDS AND BUSINESS ANALYSIS

Project Overview

This Project is to analyze sales, profitability, order trends, product performance, customer and employee's performance to support business decisions.

Business Objectives

The purpose of this project is to get the data backed insights to make better-informed business-related decisions.

How profitable is the business?

To analyze which product and categories are performing best or worst?

Which region and department are contributing most to sales?

Are sales and orders changing over time?

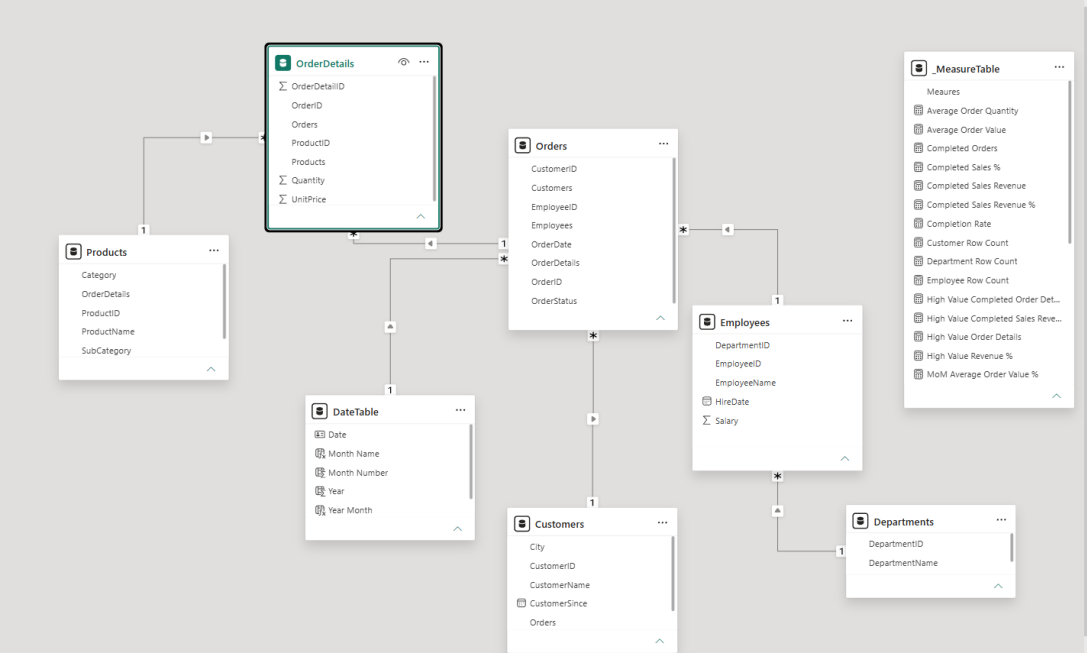
Dataset and Data Preparation

Source dataset for this project is sample data set created by me using AI. Mainly Customers, Departments, Employees, OrderDetails, Orders, Product and Date tables were created with important columns such as Customer ID, Department ID, Employee ID, Order Details ID, Order ID, Product IDs, and respective informative columns were created.

Data cleaning steps were performed like checking if all ID's are unique and not null, all the data formats of the columns were checked, Order Quantity column should be of data type Integer and not text, Date column should have Date format etc. Also checked the data set for duplicate values like in the customer information table each row should be unique.

Dimension tables were created to store descriptive information, with the primary keys used to uniquely identify records in each table. A fact table was created to store transactional data, including relevant business measures and foreign keys that link transactions to the corresponding dimension tables. This structure supports organized data analysis and reporting in Power BI.

Data Model



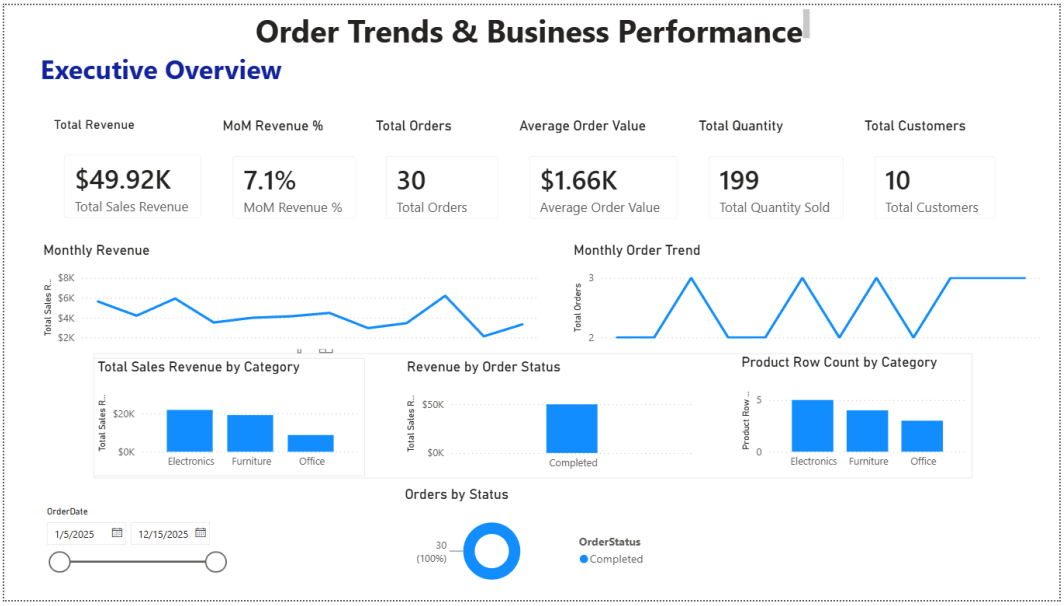
The project uses a hybrid dimensional data model with a snowflake-style structure. The Order Details table serves as the central fact table, containing order-line records and key identifiers. The Orders table connects order transactions to related information, including customers, employees, and order dates, while the Products and Departments tables provide additional context for analysis.

A separate Measure Table was created to organize DAX measures for reporting and analysis. This model supports interactive analysis of sales, profitability, order trends, product performance, customer insights, and employee performance.

6. Dashboard Pages

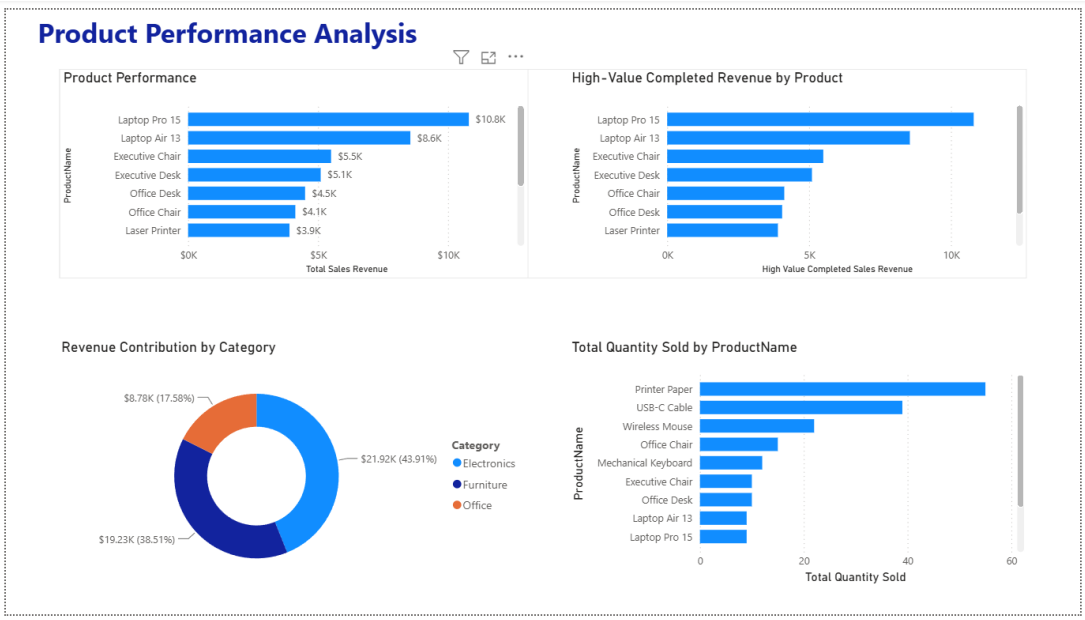
Four main report pages were developed using Power BI to provide focused insights into overall business performance, product performance, customer & employee performance. Each page included relevant KPIs, visualizations, trend charts, and interactive slicers to support data exploration and business analysis.

Executive Overview



This page provides a high-level summary of business performance through key performance indicators (KPIs) including Total Sales Revenue, Total Orders, Average Order Value, and other relevant metrics. Trend charts and category level business position.

Product Performance



This page focuses on product and category analysis using sales and profit visualization, product rankings, and category to subcategory to product drill-down functionality. These visuals help

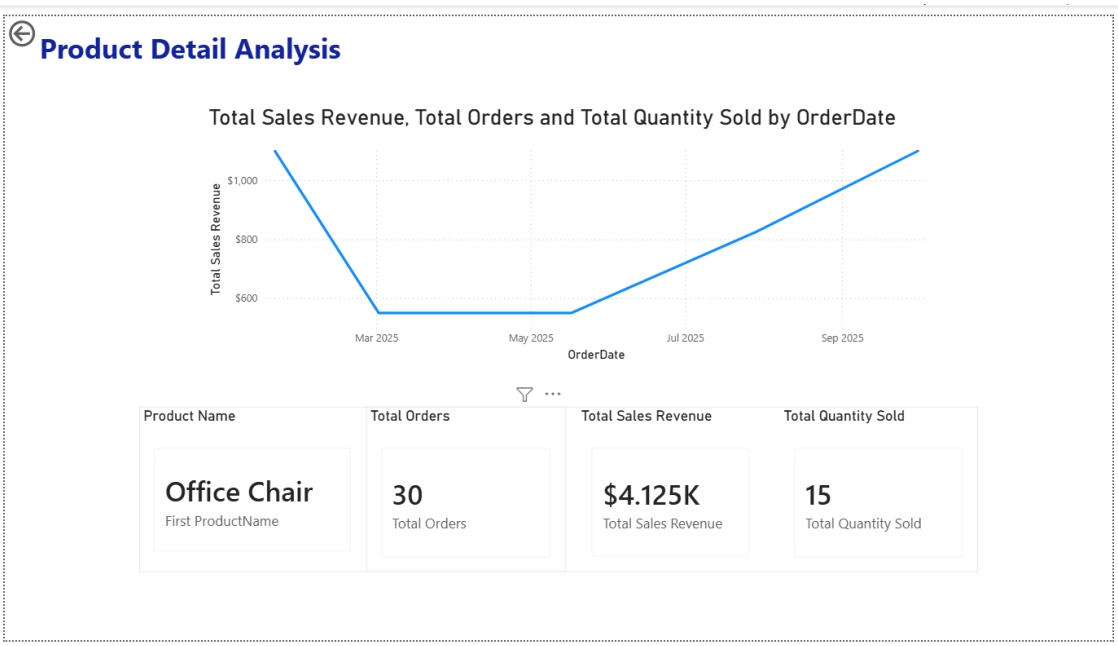
users compare product performance and identify top-performing and underperforming products.

Customer % Employee performance



Provides insights into customer contributions and employee performance using relevant KPIs and comparative visualizations. This page helps users explore customer purchasing activity and evaluate performance across employees.

Product Details



Focus of this page is to provide detailed view of individual products, including product level KPIs such as Total Sales Revenue, Total Orders, and Total Quantity sold. Trend Visualizations and interactive filtering allow users to examine product performance over the selected reporting periods.

Interactive Reporting

Slicers were incorporated into the report pages to allow users to filter the data by relevant dimensions, such as category, city, customer segment, and date. These interactive filters enable users to focus on specific products, locations, customer groups, and time periods, making the reports more flexible and useful for the business analysis.

7. Findings and Business Recommendations

Some of the actual patterns shown by the dashboard and their possible business implications.

7. Key Findings and Business Recommendations

The Power BI dashboard was designed to identify patterns in sales, profitability, order activity, product performance, customer contributions, regional performance, and employee performance.

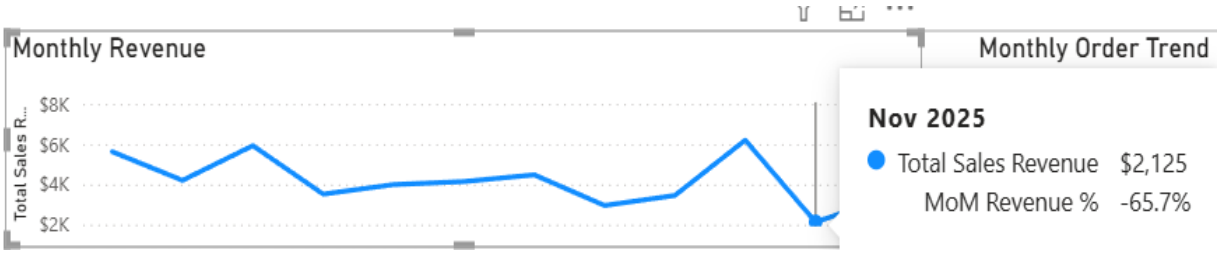
7.1 Sales and Order Trends

The monthly trend visualizations allow sales revenue and order activity to be compared over time. The analysis identifies periods of higher and lower performance and helps assess whether changes in order volume correspond to changes in revenue.

Finding:



For the month of Oct 2025 Total sales Revenue was the highest.



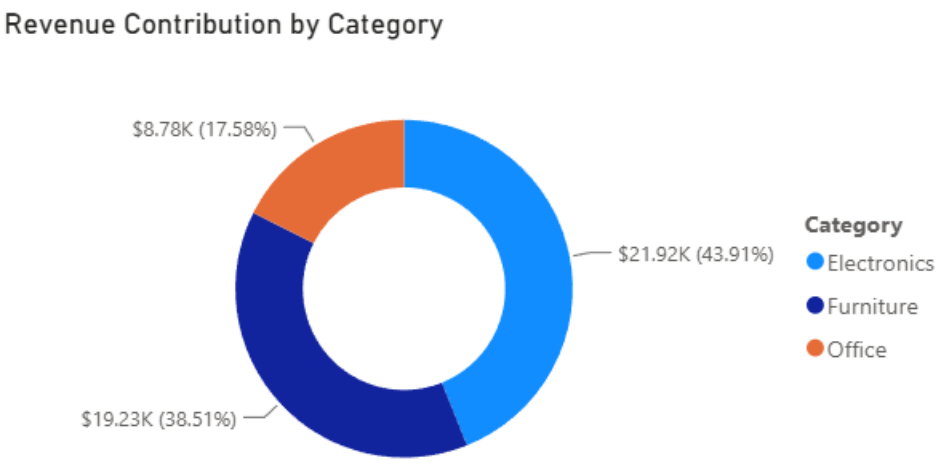
For the month of Nov 2025 was lowest.

Business implication: The sharp decline in revenue from October to November highlights the need to investigate potential changes in customer demand, order volume, product performance, or promotional activity. Understanding the factors that contribute to this decline can help the business improve demand forecasting, inventory planning, and marketing.

7.2 Product Performance

Product and category visualizations help compare sales and profitability across products and categories.

Finding:

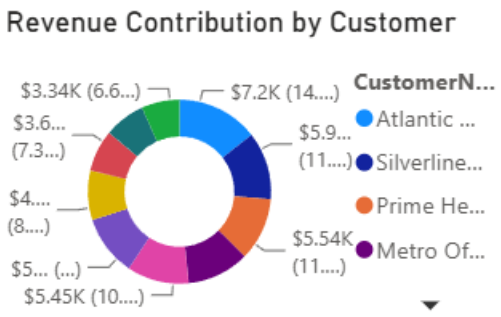


The strongest category based on actual results is electronics and weakest category is office.

Business implication: Strong-performing category may deserve closer inventory monitoring, while weaker-performing may require further investigation of demand, pricing, or profitability.

7.3 Customer Insights

Customer-segment analysis helps identify the customers or segments contributing most to overall sales.



Finding: The highest-contributing customer-segment is Atlantic Electronics, followed by Silverline Manufacturing and Prime Healthcare.

Business implication: These insights can help inform customer engagement, retention, and marketing strategies.

8. Overall Conclusion

The dashboard combines key performance indicators, trend analysis, product rankings, geographic analysis, and interactive filters in a single reporting solution. It enables users to explore business performance from multiple perspectives and supports data-informed decision-making. The findings and recommendations should be based on the actual results observed in the sample dataset.